

Sequence & Series Practice

Nine Education IIT Academy | Class 11 Mathematics | 67 Problems

Questions

1. If the sum of first n terms of an AP is $S_n = 4n^2 - n$, find its 15th term.

2. The sum of the first three terms of a GP is 21 and the sum of their squares is 189. Find the three terms.

3. The 6th term of an HP is $1/61$ and the 10th term is $1/105$. Find the first term of the HP.

4. Find the sum to infinity of the series $2+5x+8x^2+11x^3+\dots$ ($|x| < 1$).

5. Find the sum of squares of the first 18 natural numbers.

6. In an AP, the 6th term is 2 more than twice the 3rd term, and the 9th term exceeds twice the 4th term by 1. Find the first term and common difference.

7. If the 2nd term of a GP is 2 and its sum to infinity is 8, find the first term and common ratio.

8. If 4, x , 12 are in HP, find x .

9. Find the sum to infinity of the series $1+4(1/3)+7(1/3)^2+10(1/3)^3+\dots$

10. Find the sum of cubes of the first 12 natural numbers.

11. How many terms of the AP 24, 21, 18, ... must be taken so that their sum is 78?

12. Find the sum of the series $5+55+555+\dots$ to n terms.

13. Find the sum of the reciprocals of the first 6 terms of the HP whose corresponding AP has first term 2 and common difference 5.

14. Find the sum to n terms of the series $1+3x+5x^2+\dots+(2n-1)x^{n-1}$.

15. Evaluate $2^2+4^2+6^2+\dots+30^2$.

16. The sums of n terms of two APs are in the ratio $(7n+1):(4n+27)$. Find the ratio of their 11th terms.

17. The product of three consecutive terms of a GP is 512 and their sum is 28. Find the terms.

18. Insert 4 harmonic means between $1/3$ and $1/13$.

19. Find the sum to infinity of the series $3+8(1/2)+13(1/2)^2+18(1/2)^3+\dots$

20. Find the sum $1^2+3^2+5^2+\dots+29^2$.

21. If the first term of an AP is 2 and the common difference is 3, find the sum $a_1+a_3+a_5+\dots+a_{15}$ (the terms at odd positions up to the 15th term).

22. If the first term of a GP is 3 and its sum to infinity is 4, find the common ratio and the sum to infinity of the squares of its terms.

23. The 4th term of an HP is $1/16$ and its 16th term is $1/4$. Find its 64th term.

24. Find the sum of the series $2+22+222+\dots$ to n terms.

25. If the sum of cubes of the first n natural numbers is 2025, find n .

26. Three numbers are in AP such that their sum is 24 and the sum of their squares is 200. Find the numbers.

27. Find the number of terms in the GP 3, 6, 12, ..., 384.

28. The 5th and 10th terms of an HP are $\frac{1}{9}$ and $\frac{1}{19}$ respectively. Find its 20th term.

29. If $S = x + 3x^2 + 5x^3 + \dots$ to infinity, find the value of S at $x = \frac{1}{2}$.

30. Evaluate the sum of $r^2(r-1)$ for $r = 1$ to 8.

31. If the p th, q th, and r th terms of an AP are x , y , z respectively, find the value of $x(q-r)+y(r-p)+z(p-q)$.

32. The sum to infinity of a GP is 3 times the sum of its even-position terms. Find the common ratio.

33. Find the harmonic mean of the roots of the equation $x^2 - 10x + 9 = 0$.

34. Find the sum to infinity of the series $\sum n \cdot \left(\frac{1}{2}\right)^{n+1}$ for $n = 1$ to ∞ .

35. If the sum of squares of the first n natural numbers is 3311, find n .

36. The sum of the first 10 terms of an AP is 145 and the sum of the next 10 terms (11th to 20th) is 445. Find the common difference.

37. Insert 3 geometric means between 1 and 256.

38. The AM and HM of two numbers are 25 and 16 respectively. Find the numbers.

39. Find the sum to infinity of the series $4 + 9\left(\frac{2}{5}\right) + 14\left(\frac{2}{5}\right)^2 + 19\left(\frac{2}{5}\right)^3 + \dots$

40. Evaluate $1^3 + 3^3 + 5^3 + \dots + 15^3$.

41. Find the sum of all integers between 100 and 200 which are divisible by 9.

42. Three numbers whose sum is 70 are in GP. If the extreme terms are each multiplied by 4 and the middle term by 5, the results form an AP. Find the numbers.

43. The first two terms of an HP are $\frac{1}{5}$ and $\frac{1}{9}$. Find its 15th term, and state whether $\frac{1}{50}$ is a term of this HP.

44. Find the sum of the first 4 terms of the series $1 + 2(3) + 3(9) + 4(27) + \dots$

45. Evaluate the sum of $(r^3 + 2r^2)$ for $r = 1$ to 6.

46. The interior angles of a polygon are in AP. The smallest angle is 120° and the common difference is 5° . If the largest angle is 160° , find the number of sides of the polygon.

47. The 3rd term of a GP equals the square of its first term, and the 5th term is 8 times the 3rd term. Find the common ratio.

48. Find the 8th term of the HP $\frac{1}{2}, \frac{1}{5}, \frac{1}{8}, \dots$

49. Find the sum to infinity of the series $1 + 6\left(\frac{1}{3}\right) + 11\left(\frac{1}{3}\right)^2 + 16\left(\frac{1}{3}\right)^3 + \dots$

50. The sum of squares of the first n natural numbers exceeds the sum of the first n natural numbers by 240. Find n .

51. The sum of the first 12 terms of an AP is 168 and its first term is 3. Find the common difference.

52. The first term of a GP exceeds the second term by 2, and the sum to infinity is 50. Find the first term and common ratio.

53. The 7th term of an HP is $1/10$ and the 12th term is $1/20$. Find the 3rd term of the HP.

54. Find the sum to infinity of the series $1+3(0.6)+5(0.6)^2+7(0.6)^3+\dots$

55. Find the sum of squares of the first 16 natural numbers.

56. The 3rd term of an AP is 7, and the 7th term exceeds three times the 3rd term by 2. Find the common difference and the 10th term.

57. The first term of a GP is 2 and its 4th term is 54. Find the sum of its first 5 terms.

58. The harmonic mean of two numbers is 4, and their arithmetic mean is 4.5. Find the two numbers.

59. Find the sum to n terms of the series $2+5x+8x^2+\dots+(3n-1)x^{n-1}$ at $x=1$.

60. Evaluate $3^3+6^3+9^3+\dots+30^3$.

61. Find the sum of all three-digit natural numbers which leave remainder 3 when divided by 5.

62. If $x, x+3, x+9$ are the first three terms of a GP, find x and the common ratio.

63. Numbers a, b, c are in HP such that $a=6$ and $c=3$, with b as their harmonic mean. Find b .

64. Find the sum to infinity of the series $5+8(1/4)+11(1/4)^2+14(1/4)^3+\dots$

65. If the sum of the first n natural numbers equals the sum of squares of the first 6 natural numbers, find n .

66. The sum of first p terms of an AP is q , and the sum of its first q terms is p ($p \neq q$). Find the sum of its first $(p+q)$ terms.

67. Find the value of $[1^3+2^3+\dots+20^3] - [1^2+2^2+\dots+20^2]$.

Answer Key

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|---|------------------------------|--------------------------|
| 1. 115 | 23. $1/64$ | 45. 623 |
| 2. 3, 6, 12 | 24. $(2/81)(10^{n+1}-9n-10)$ | 46. $n=9$ |
| 3. $1/6$ | 25. $n=9$ | 47. $r=\pm 2\sqrt{2}$ |
| 4. $(2+x)/(1-x)^2$ | 26. 6, 8, 10 | 48. $1/23$ |
| 5. 2109 | 27. 8 | 49. $21/4$ |
| 6. $a=-3, d=-1$ | 28. $1/39$ | 50. $n=9$ |
| 7. $a=4, r=1/2$ | 29. 3 | 51. $d=2$ |
| 8. $x=6$ | 30. 1092 | 52. $a=10, r=4/5$ |
| 9. $15/4$ | 31. 0 | 53. $1/2$ |
| 10. 6084 | 32. $1/2$ | 54. 10 |
| 11. $n=4$ or 13 | 33. $9/5$ | 55. 1496 |
| 12. $(5/81)(10^{n+1}-9n-10)$ | 34. 1 | 56. $d=4$, 10th term=35 |
| 13. 87 | 35. $n=21$ | 57. 242 |
| 14. $[1-(2n-1)x^n]/(1-x) + 2x(1-x^{n-1})/(1-x)^2$ | 36. 3 | 58. 3, 6 |
| 15. 4960 | 37. 4, 16, 64 | 59. $n(3n+1)/2$ |
| 16. 4:3 | 38. 10, 40 | 60. 81675 |
| 17. 4, 8, 16 | 39. $110/9$ | 61. 99090 |
| 18. $1/5, 1/7, 1/9, 1/11$ | 40. 8128 | 62. $x=3, r=2$ |
| 19. 16 | 41. 1683 | 63. $b=4$ |
| 20. 4495 | 42. 10, 20, 40 | 64. 8 |
| 21. 184 | 43. $1/61$; No | 65. $n=13$ |
| 22. $r=1/4$, sum of squares=48/5 | 44. 142 | 66. $-(p+q)$ |
| | | 67. 41230 |